Ablation studies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask what remains fixed and which failures the total hides.

Teach controlled component removal without additive or universal attribution.

Keep one habit: Model net drops → inspect unauthorized-action boundary → show unknown joint cell → contrast filter/model confound → assess independent totals.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **ablation** studies

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of multiple comparisons

Reference example, not the only acceptable wording: Teacher explanation: Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

For this exercise use Bonferroni:family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of effect size and practical importance

Reference example, not the only acceptable wording: Teacher explanation: Effect size describes magnitude on a stated scale. Practical importance depends on task requirements, not merely a nonzero difference.

For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of confounders and controls

Reference example, not the only acceptable wording: Teacher explanation: A confounder changes alongside the intended factor and can offer another explanation. Controls aim to isolate the proposed **mechanism**.

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Pictured end goal

Explain **ablation** studies using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Multiple comparisons

Effect size and practical importance

Confounders and controls

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Ablation** removes a component to investigate its contribution under a controlled comparison. **Interaction**s can make contributions depend on other components.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **ablation** studies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach ablations as controlled conditional comparisons

> Inspect critical failures and the missing joint-removal cell alongside scores.

Definitions and full contract

Ablation removes a component to investigate its contribution under a controlled comparison. **Interaction**s can make contributions depend on other components.

Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.

Comparison item	Complete constructed contract
Common evaluation	Same ten cases, same scoring and all other settings fixed
Full system	Retrieval and validation present; 8/10 passes
Remove retrieval only	Validation remains; 5/10 passes
Remove validation only	Retrieval remains; 7/10 passes; an unauthorized action is stipulated
Remove both	Not supplied; do not invent this outcome
Reported effects	Full minus ablated pass count and percentage-point difference
Case identities	Not supplied; net differences do not identify which cases flipped

Comparison item	Complete constructed contract
Scope	Supplied fictional summaries, not actual model runs or population causal estimates

Main ledger and checked teacher reasoning

Condition	Pass count/rate	Net drop from full	Interpretation
Full system	8/10 = 80%	Reference	Both components present
Without retrieval	5/10 = 50%	3 passes; 30 percentage points	Conditional removal effect in this configuration
Without validation	7/10 = 70%	1 pass; 10 percentage points	Also permits a stipulated unauthorized action
Without both	Unknown	Unknown	Cannot calculate joint effect or interaction
Changed case identities	Unknown	Net count is not a flip ledger	Inspect per-case records if collected

Main: retrieval removal drops 8 to 5 passes, three passes or thirty percentage points. Validation removal drops 8 to 7, one pass or ten points, and permits a stipulated unauthorized action. Score magnitude alone does not capture that boundary. Joint removal is unknown; compatible hypothetical neither counts 4 or 6 yield interaction contrasts 0 or 2.

Separate filter example: 8/10 to 6/10 is a net two passes, twenty percentage points or a 25% relative decrease. Exact changed cases are unknown. Selected independent totals 12/15, 9/15 and 11/15 give drops 3/15 = 20 points and 1/15 ≈ 6.667 points, conditional on the remaining components. No real model execution, significance or universal causal contribution follows.

Ablation studies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action.	Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.	Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone. Qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action.

Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Interaction and severity probes

Ablation audit	Checked interpretation
Removing validation drops only one pass	Do not dismiss the stipulated unauthorized action
Adding the two observed drops	Does not determine the unobserved joint removal
Both-removal count hypothetically 4	Interaction contrast 8 – 7 – 5 + 4 = 0
Both-removal count hypothetically 6	Interaction contrast 8 – 7 – 5 + 6 = 2; also compatible with supplied three cells
Hypothetical counterexamples	Neither is an observed result; they demonstrate missing information
Net score falls by two	Per-case gains and losses may offset; exact flips unknown
Filter and model both change	Alternative explanations prevent filter-only attribution
All settings said to be fixed	An exercise assumption; actual study would need recorded settings and outputs

Separate filter inputs and native main grid

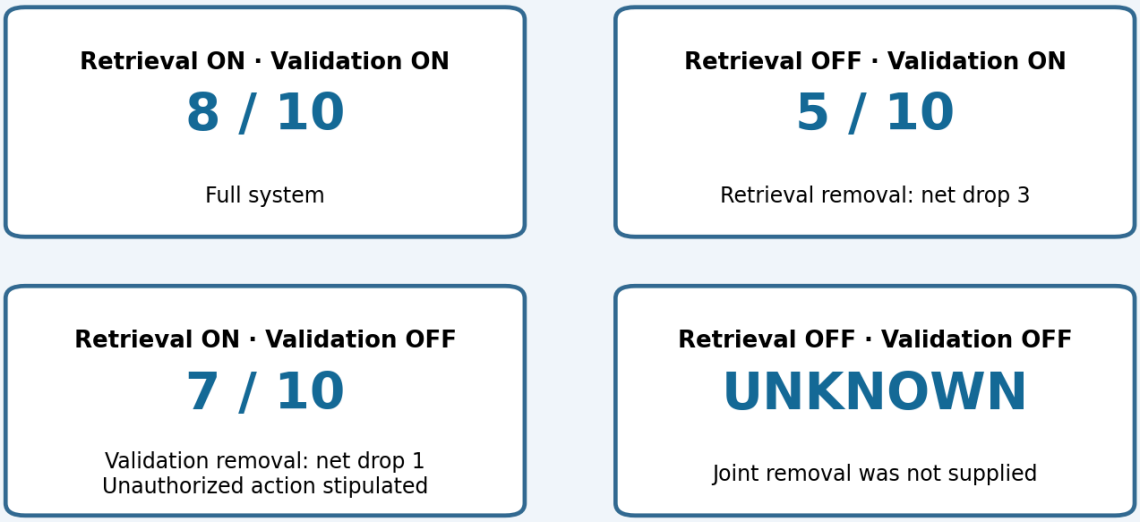
Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter’s contribution.

Full pipeline: 8/10 correct. Filter removed: 6/10 on the same cases, with all other settings fixed.

The main grid separates retrieval and validation. Its two observed net score drops are three and one passes, but their sum does not establish the result of removing both. The separate picture investigates a relevance filter instead: full pipeline 8/10 versus filter removed 6/10, a net drop of two passes or twenty percentage points.

The aggregate counts do not show whether exactly two cases changed; gains and losses could offset. If the model version also changes, the comparison cannot isolate the filter. Neither picture records a real model execution or proves a universal contribution.

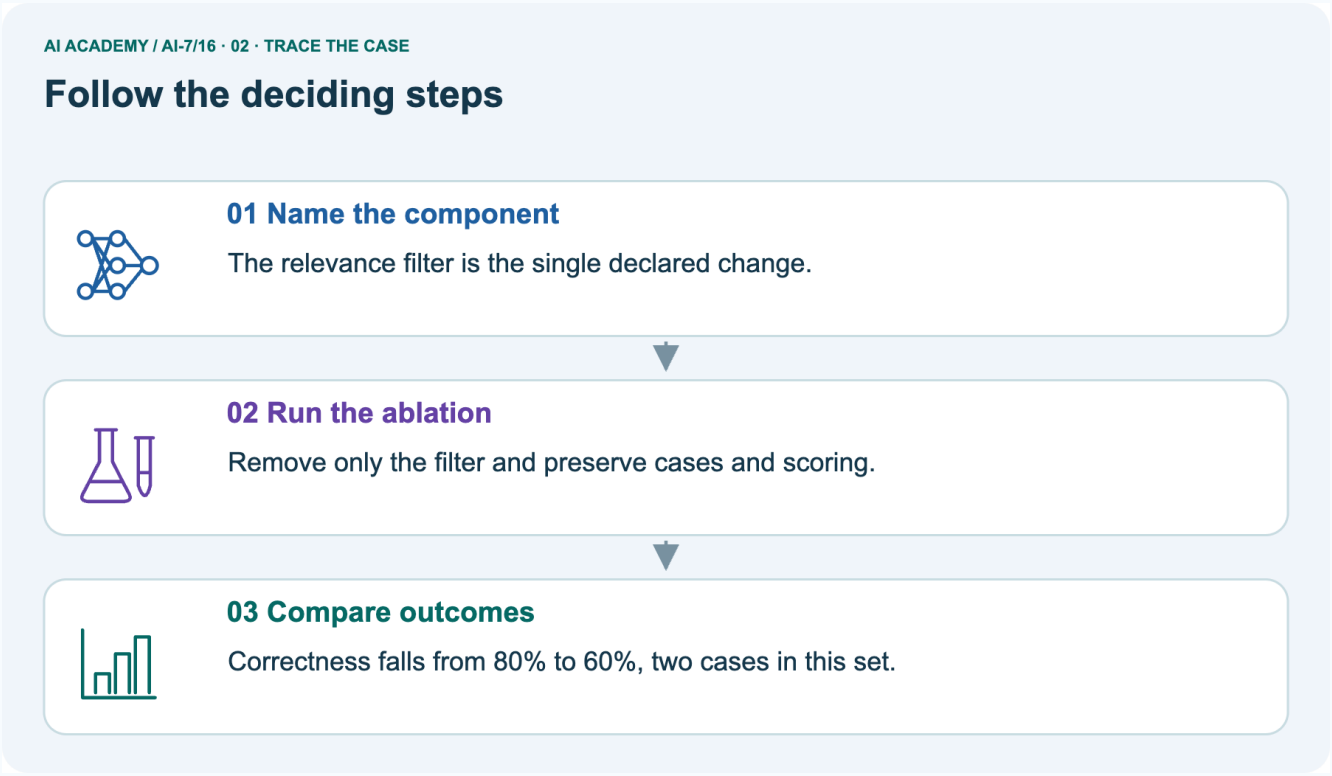
Ablation contributions depend on what remains



Same ten cases and scoring; other settings held fixed by the constructed exercise.
The unknown cell prevents calculating the interaction; net counts do not identify case flips.

Separate filter picture	Calculation or evidence limit
Full pipeline	8/10 = 80%
Filter removed only	6/10 = 60%
Net difference	8 – 6 = 2 passes; 80% – 60% = 20 percentage points
Relative decrease from full	2/8 = 25%; differs from 20 percentage points
Changed model as well	Two factors vary; filter-only attribution is lost
Which cases changed	Not supplied; net two is not proof of exactly two flips
General conclusion	Conditional descriptive comparison; no universal benefit or significance established

Picture trace and changed-model explanation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name the component	The relevance filter is the single declared change.
2. Run the ablation	Remove only the filter and preserve cases and scoring.
3. Compare outcomes	Correctness falls from 80% to 60%, two cases in this set.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Interactions and small-sample variability remain; this illustrative difference is not a universal causal estimate.

Change: Remove the filter and also change the model version.

Prediction: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Ablation	Removing or disabling a component in a comparison
Conditional contribution	An effect given the other components
Interaction	An effect depending on another factor's setting

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

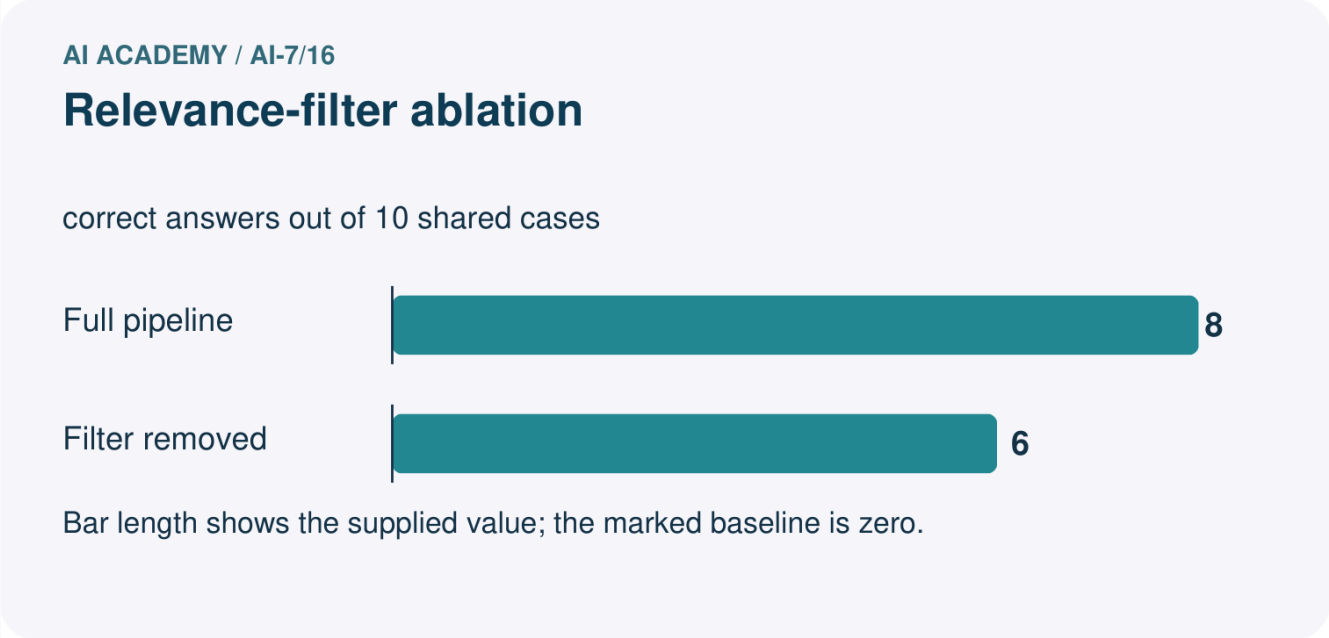
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Ablation studies

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/16

Ablation studies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the component

The relevance filter is the single declared change.



2. Run the ablation

Remove only the filter and preserve cases and scoring.



3. Compare outcomes

Correctness falls from 80% to 60%, two cases in this set.

CHANGE ONE THING

Remove the filter and also change the model version.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Ablation** studies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the filter **ablation**'s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

The relevance filter is the single declared change. Remove only the filter and preserve cases and scoring. Correctness falls from **80%** to **60%**, two cases in this set. Changed case: The comparison no longer isolates the filter.

Two simultaneous changes provide alternative explanations.

Changed-case reasoning: The comparison no longer isolates the filter. Two simultaneous changes provide alternative explanations.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main net drops are three and one passes, with a critical unauthorized-action failure in the latter condition. Missing joint-removal results prevent calculating interaction. Separate filter drop is two passes or twenty percentage points. Independent totals 12/15, 9/15 and 11/15 give three-pass and one-pass drops: twenty and approximately 6.667 percentage points.

Contributions remain conditional on other components and are not universally additive.

Diagnose the current misconceptions

The worked result establishes every future case: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Full 12/15, without A 9/15, without B 11/15. Compute sample drops and state **interaction** limit.

Teacher explanation: Drops 3/15=20 percentage points and 1/15≈6.667 points. Effects are conditional on remaining components; not additive universal contributions.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Ablation** removes a component to investigate its contribution under a controlled comparison. **Interaction**s can make contributions depend on other components.

Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.

W2 Trace the worked case

Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

W3 Audit the limit

A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone. Qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Full 12/15, without A 9/15, without B 11/15. Compute sample drops and state **interaction** limit.

Teacher explanation: Drops 3/15=20 percentage points and 1/15≈6.667 points. Effects are conditional on remaining components; not additive universal contributions.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Drops 3/15=20 percentage points and 1/15≈6.667 points. Effects are conditional on remaining components; not additive universal contributions. Require intermediate reasoning and the stated scope. Fresh retry: Full 9/12, without A 6/12, without B 8/12. Expected reasoning: Drops 25 and 8.333 percentage points; keep identical cases and inspect individual failures. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **ablation** studies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **ablation** studies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Sensitivity analysis. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/17, ai-7/19, ai-7/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **ablation** studies

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

Check every learner before discussion: Use the worked case for **Ablation** studies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Remove the filter and also change the model version. Require a **prediction** and a reason before comparing. Expected result: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

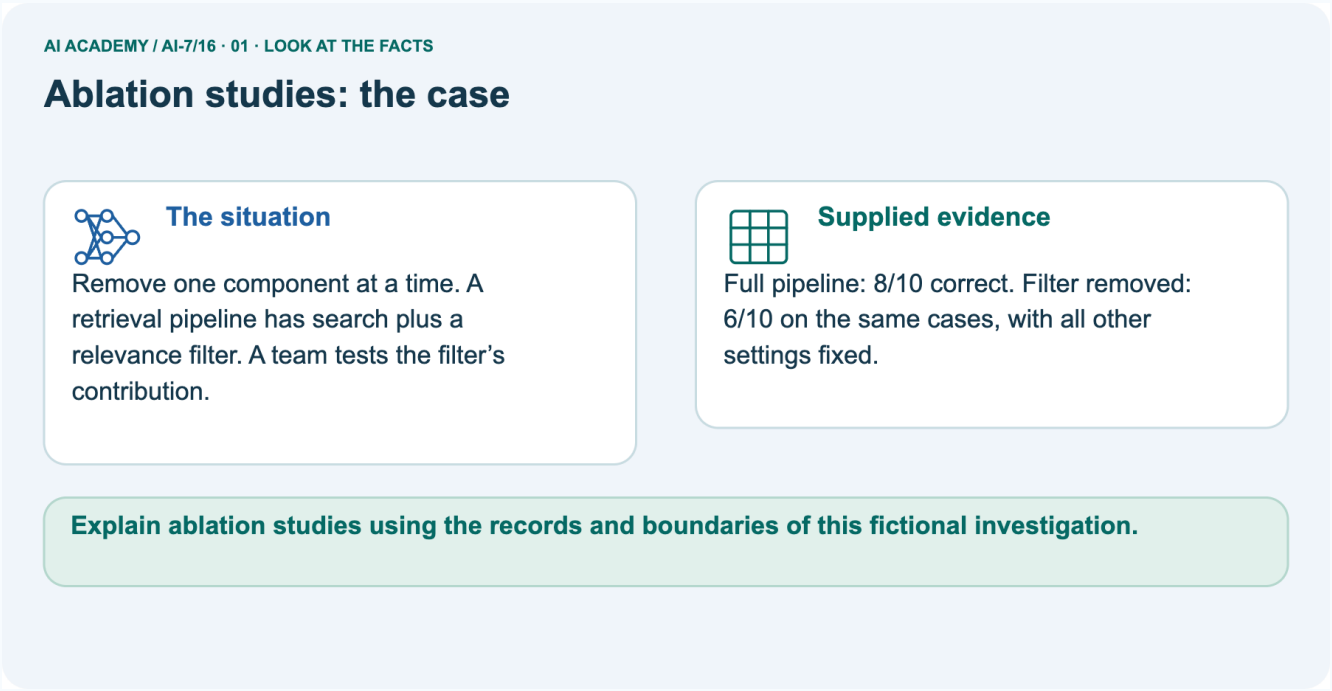
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Ablation studies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

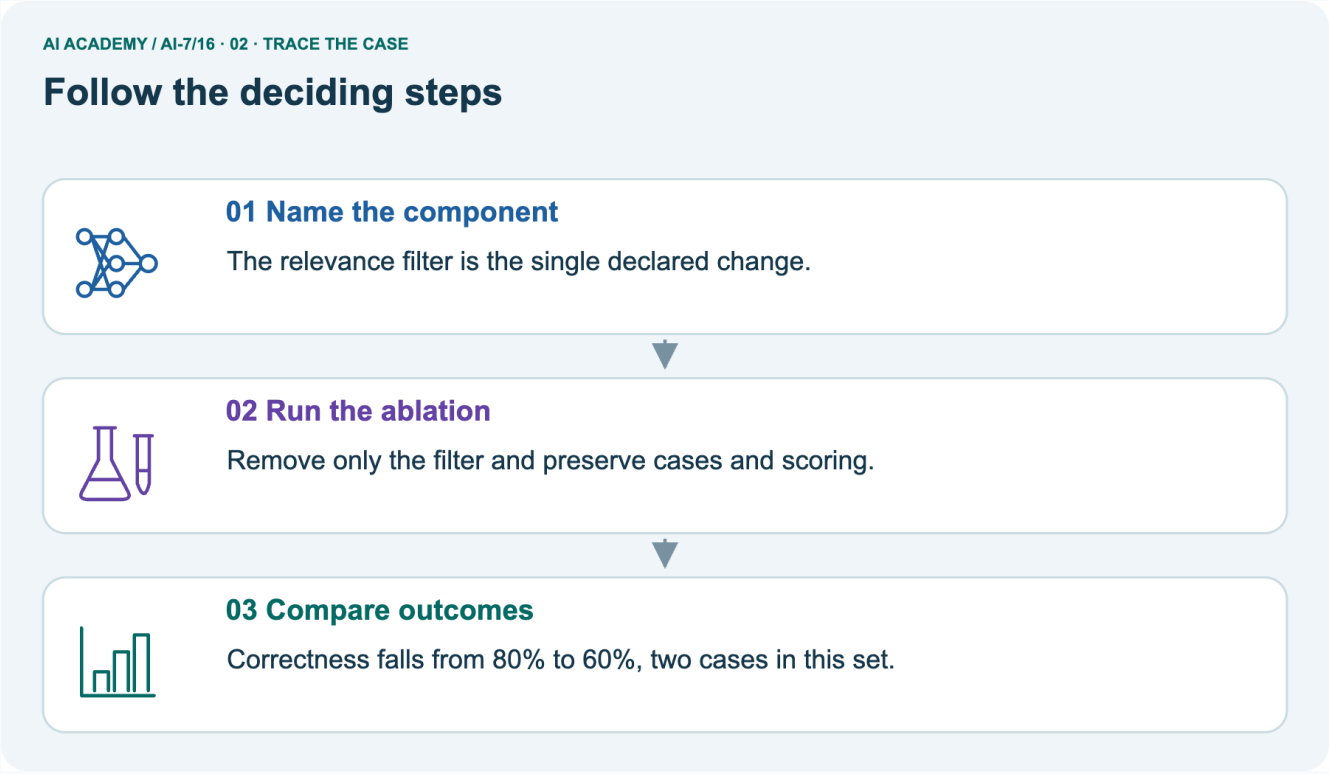
Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution. Supplied facts: Full pipeline: 8/10 correct.

Filter removed: 6/10 on the same cases, with all other settings fixed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



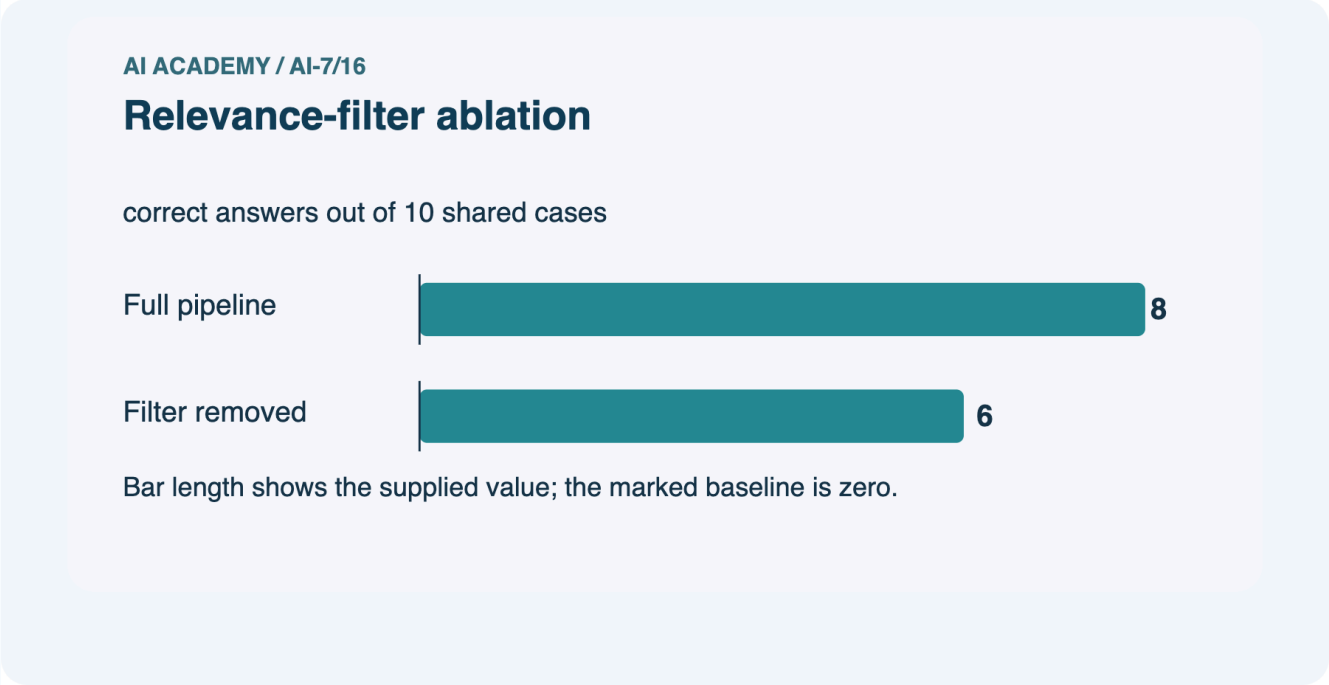
Follow the labelled picture with the matching explanation. Open the vector diagram

Name the component: The relevance filter is the single declared change. → Run the **ablation**: Remove only the filter and preserve cases and scoring. → Compare outcomes: Correctness falls from **80%** to **60%**, two cases in this set.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Relevance-filter ablation



Follow the labelled picture with the matching explanation. Open the vector diagram

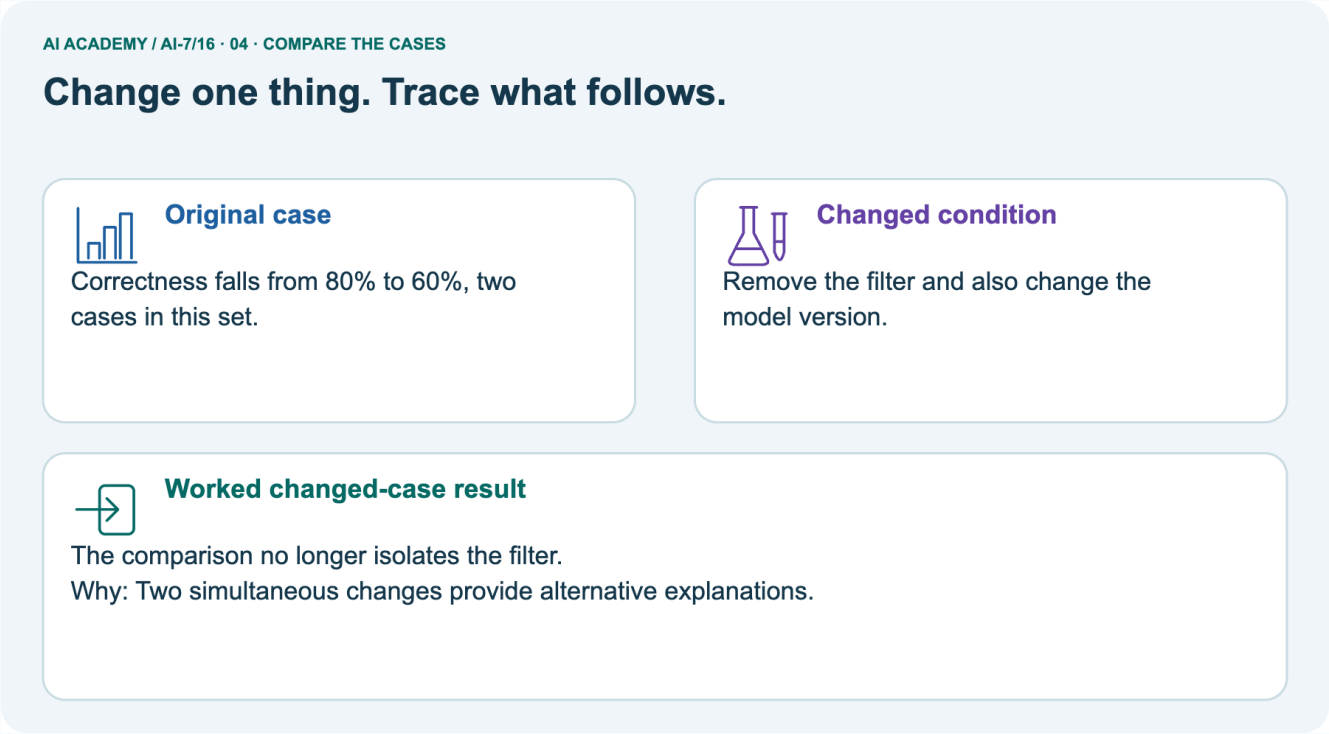
Relevance-filter **ablation**. Units: correct answers out of 10 shared cases. Full pipeline: 8. Filter removed: 6.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correctness falls from **80%** to **60%**, two cases in this set. Changed condition: Remove the filter and also change the model version. Expected result: The comparison no longer isolates the filter.

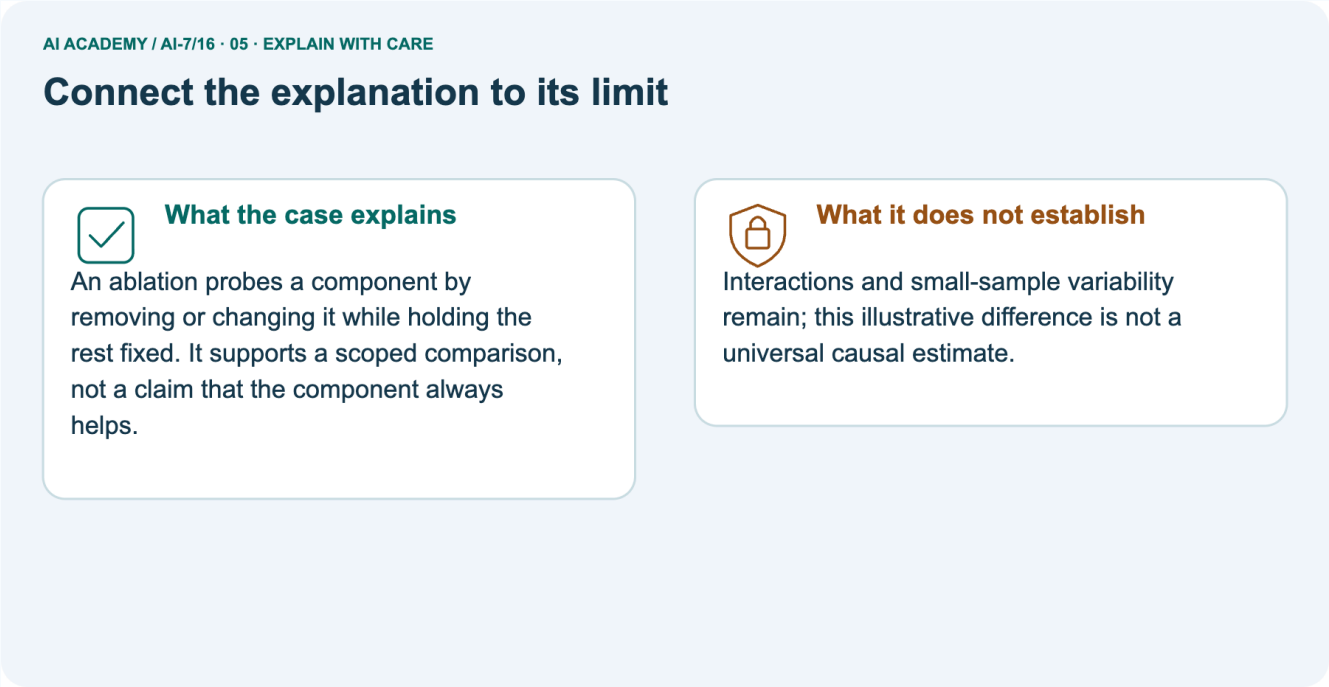
Reason: Two simultaneous changes provide alternative explanations.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Two simultaneous changes provide alternative explanations.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

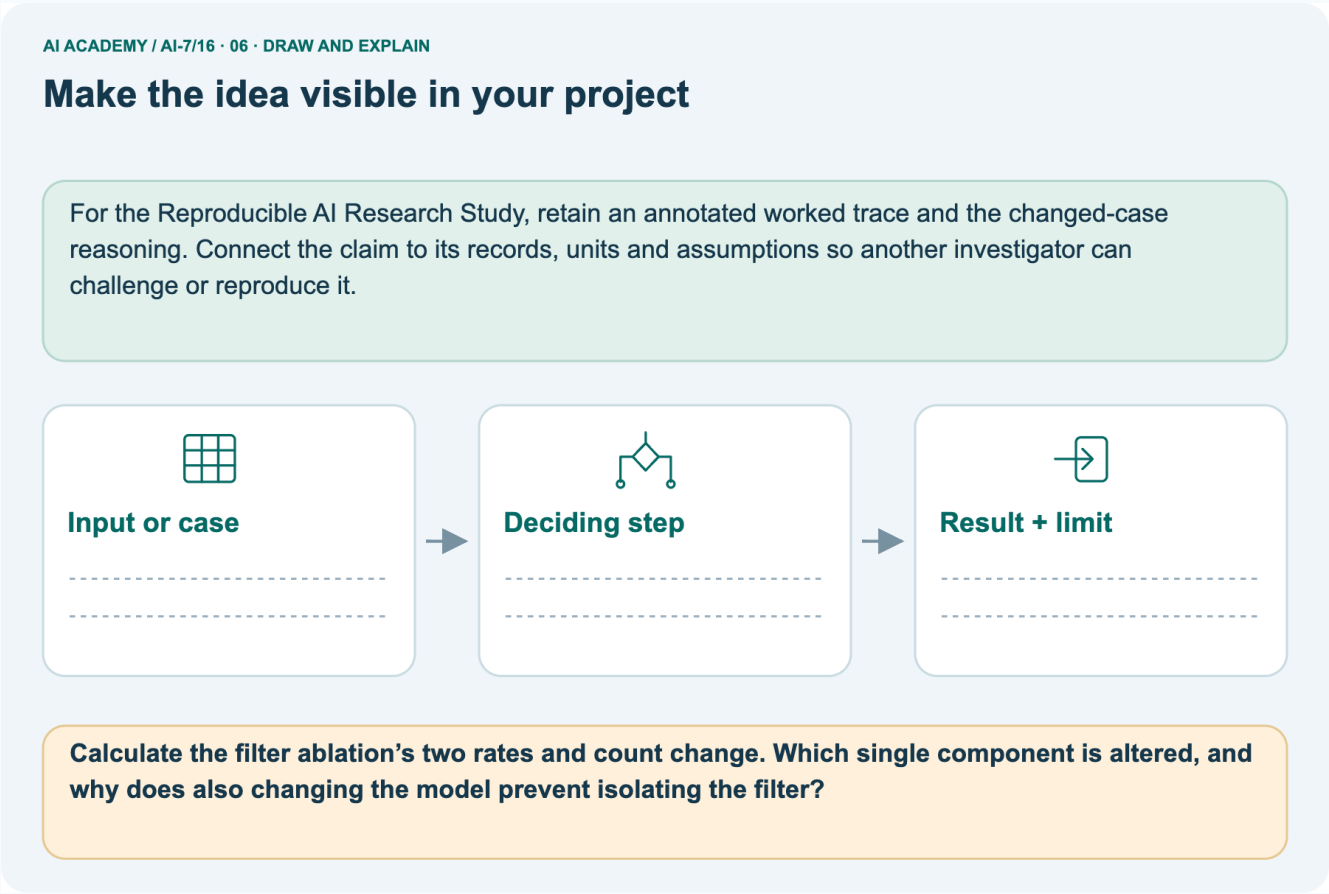
Limit: **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the filter **ablation**'s two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The relevance filter is the single declared change. Remove only the filter and preserve cases and scoring. Correctness falls from **80%** to **60%**, two cases in this set.

Changed case: The comparison no longer isolates the filter. Two simultaneous changes provide alternative explanations.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Ablation studies: the case

02. Follow the deciding steps

03. Relevance-filter ablation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.